# Leverage Direction Matters

Cross-asset GARCH asymmetry, volatility-targeting behavior, and delayed Asia spillover in one framework

PANEL A

**Leverage direction is not univer al**

## Gold flips sign

fear-rally episodes show inverted leverage

**t = -5.79**

93% of rolling gamma estimates are negativ

Rule of thumb:

use GJR-GARCH only when

**gamma > 0.10**

12/12 DM comparisons classified correctly

PANEL B

**Volatility targeting has two cha els**

## Signal channel

Within equities, gamma predicts whether V behaves like trend-following or contrarian

**rho = 0.886**

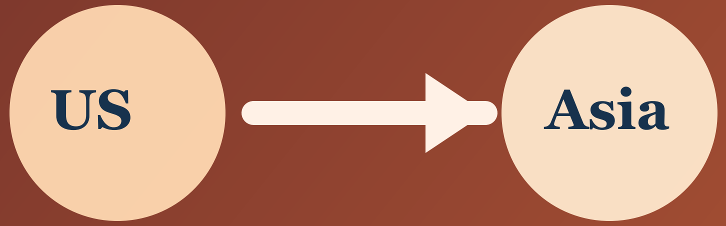
## Insurance channel

Drawdown reduction is broad-based and tie base volatility rather than asymmetry sign

**rho = 0.944**

PANEL C

**U.S. close predicts Asia next day**




One-day delayed spillover survives Harvey multiple-testing threshold

**6 markets**

with t > 3.0 after costs

Practical implication: local closing-time allocation rules can remain simple and trad

Takeaway: asymmetry sign tells you when extra GARCH complexity matters; volatility targeting mostly earns its keep through universal drawdown control, with a separate delaye